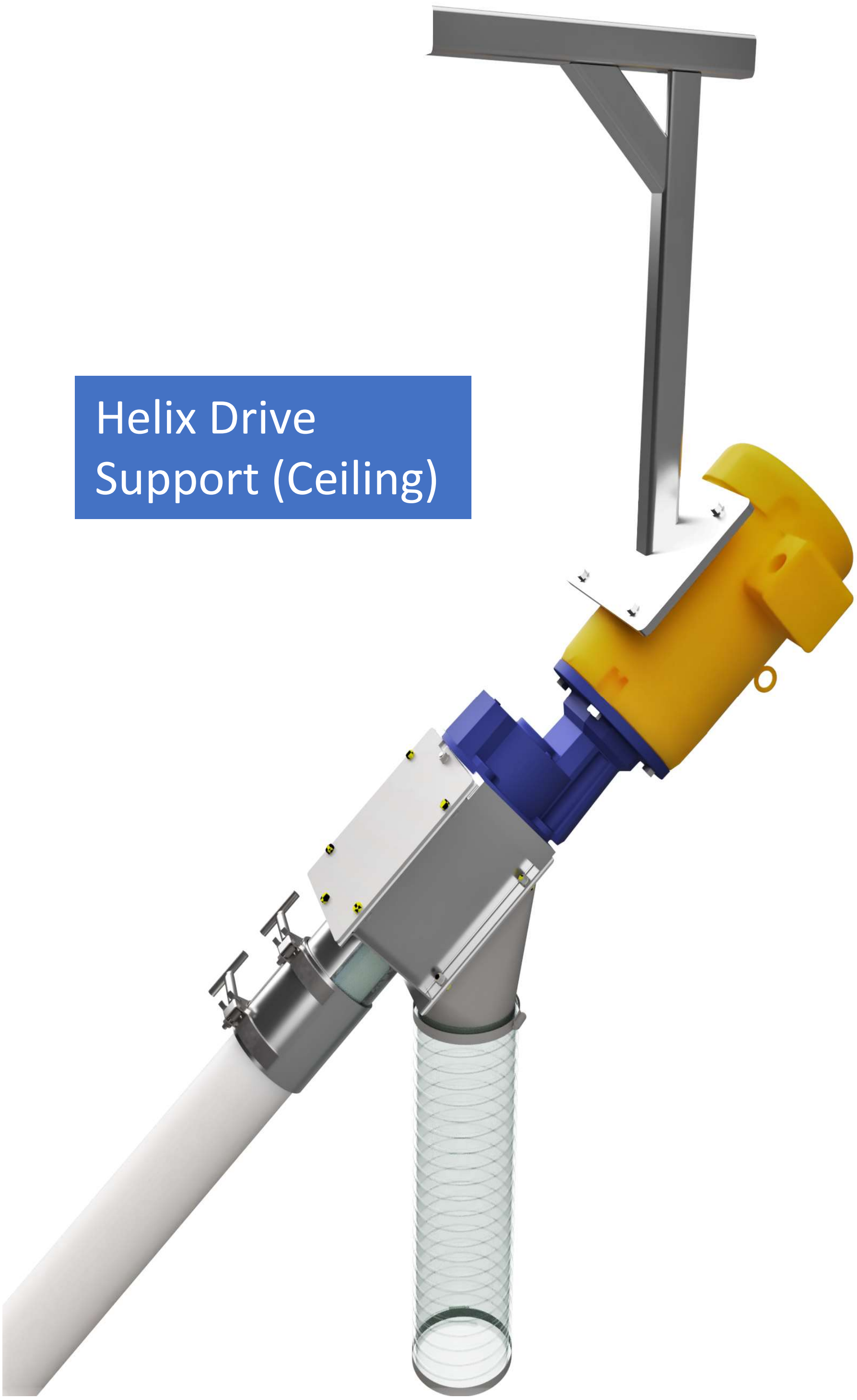
7.0 Installation Pre-Checks

1.	Upon arrival of the unit, inspect all components to be sure that there has been no damage in shipment.
2.	Determine where to mount the drive assembly and position it so it will properly discharge material into the vessel being filled.
NOTICE	If the vessel being filled contains vapors that could rise and create moisture or back pressure in the drive box, then some type of isolation device (slide gate, butterfly valve, etc..) should be installed between the helix discharge and the vessel being filled.
3.	Place the Helix™ hopper into the desired position and observe the path that the casing/auger must travel to reach drive assembly.
4.	Make sure there is enough room at the back of the hopper that the auger can be pulled out through the end cap or enough room to be able to slide the hopper back out of the way to remove the auger.
5.	The conveyor path can now be finalized, and drive support requirements determined.
6.	Note any distances or angles that differ from the Hapman supplied top level print. As the auger and casing may need to be modified to account for these changes.

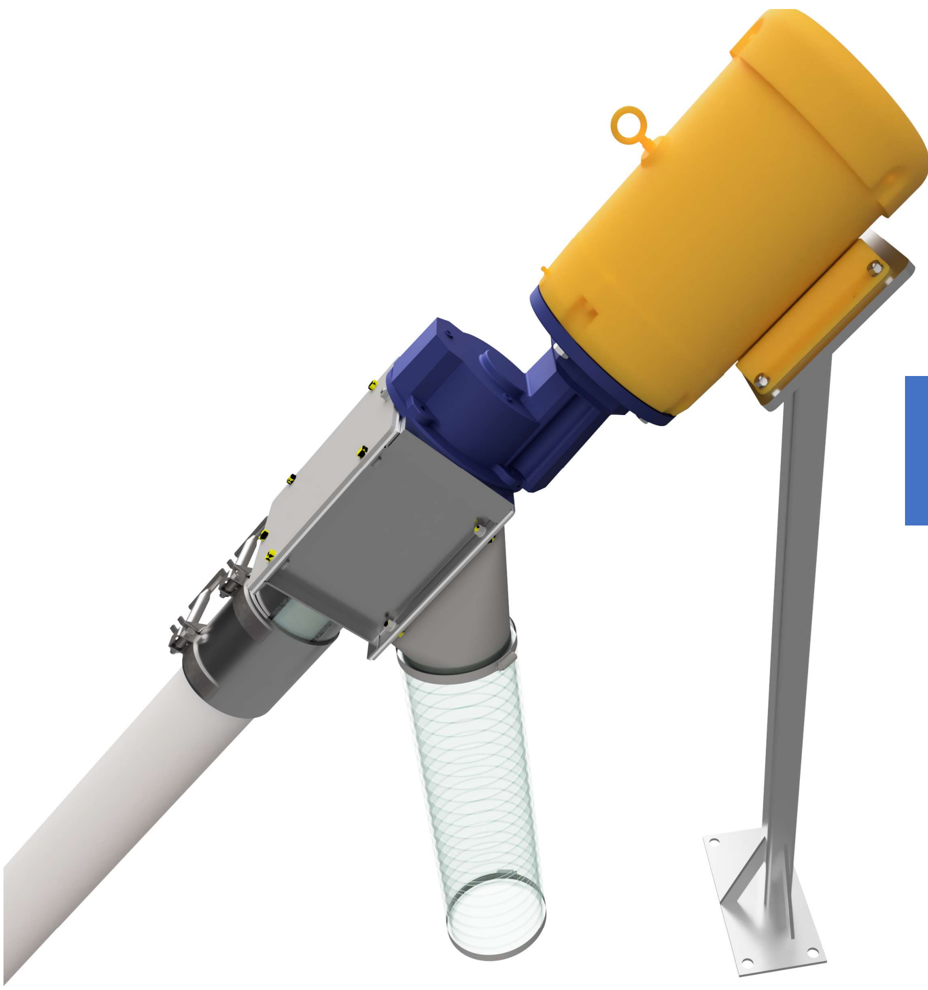
8.0 Installation: Drive Assembly	
1.	Complete all the installation pre-checks from the previous page.
2.	Install the drive assembly.
3.	Verify that the gear reducer has the breather vent installed at the highest point on the gear reduce.
NOTICE	*The drive assembly must be securely supported.*
NOTICE	*If overhead support is not available, a floor mounted mast should be fabricated or purchased from Hapman before continuing the installation.* ***(Examples of support stands shown on next two pages)***
NOTICE	*The use of all thread to hang the drive assembly is not recommended.*

8.0 Installation: Drive Assembly Motor Support Examples

Helix Drive Support (Ceiling)



8.0 Installation (Drive Assembly) Motor Support Examples

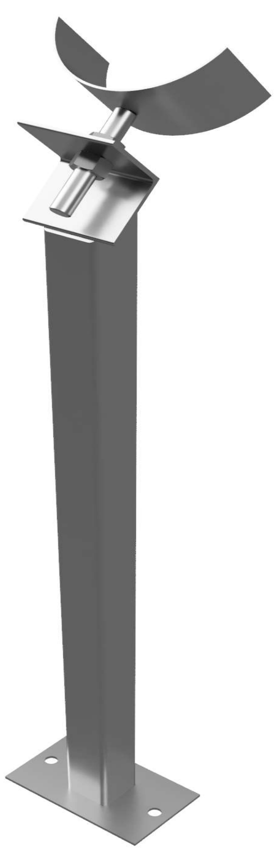


Helix Drive Support (Floor)

8.0 Installation: Helix Casing

Below are the instructions for installation of the 250 and 300 series Helix™ casing assemblies.
If it is necessary for the auger/casing to be curved, please reference the charts on *pages 28-29* before installing. Ensure that the curve being implemented is within Hapman's specifications.

NOTICE	*On long conveyor runs, it may be necessary to support the casing. Additional support will minimize sag, reduce vibration and strain on connections.* ***(Examples of casing supports shown below)***
NOTICE	*The casing must not exert lateral force on drive assembly or in-feed hopper because of curves in conveyor casing.*
NOTICE	*Isolate casing from sharp or protruding objects that may cause casing to collapse.*
NOTICE	*Casing exerting excessive strain on drive assembly or in-feed hopper may change alignment of those components.*

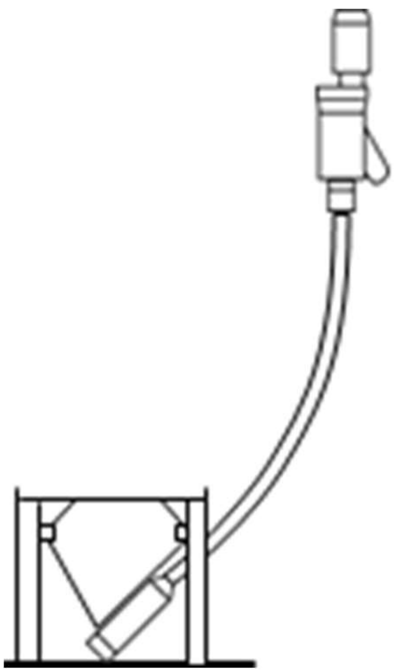


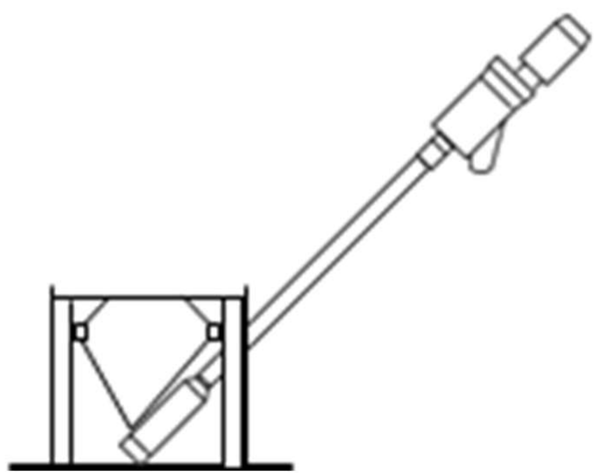
8.0 Installation: Helix Casing

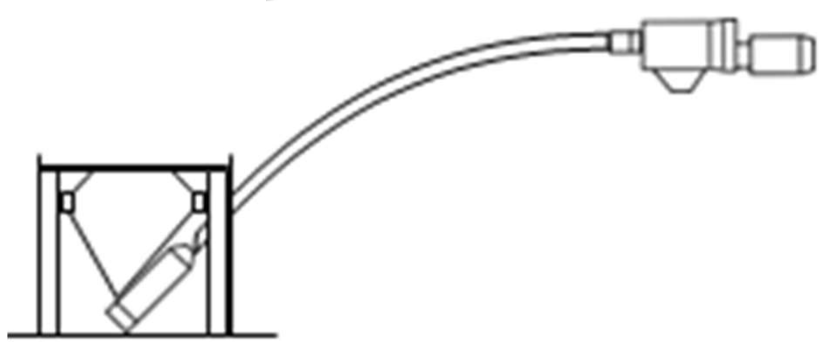
Below are examples of the correct ways to install a Helix casing.

NOTICE	*If it is necessary for the auger/casing to be curved, maintain a smooth and generous bend radius.*

Please reference our Helix casing bend charts on pages 28-29 of this manual before installing.







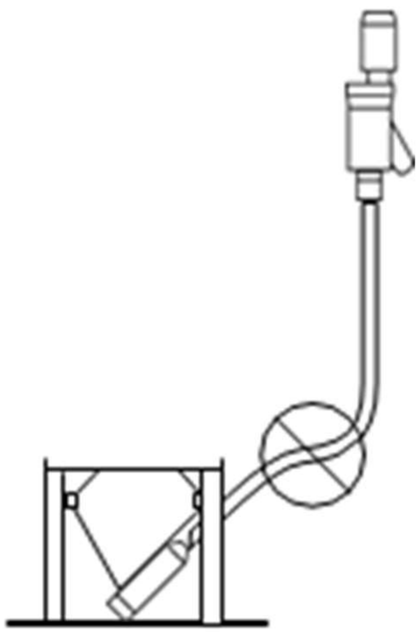
8.0 Installation: Helix Casing

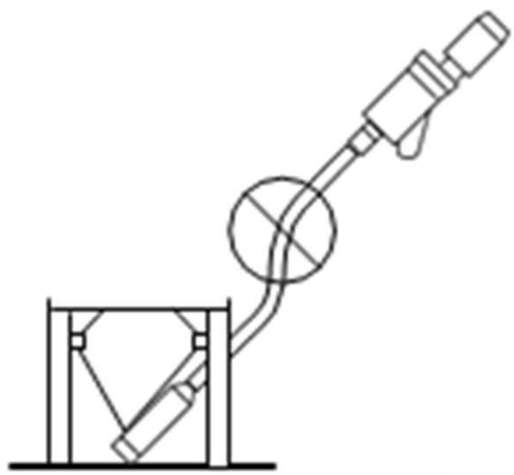
Below are examples of ***improper*** *casing installations.*

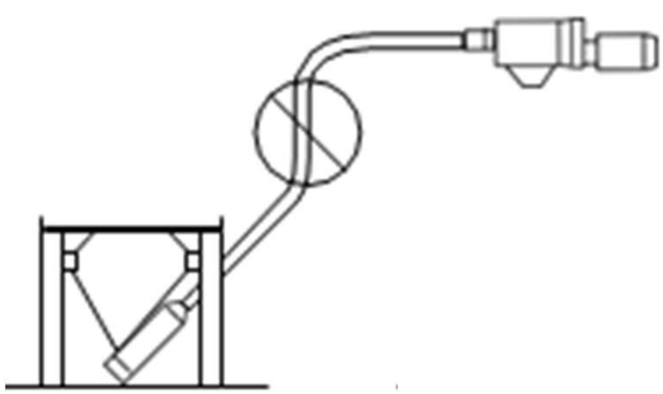
Common mistakes made during a casing installation are:

- Putting more than one bend in the casing
- Having a bend with too tight of a radius

Putting a tight radius bend or more than one bend in the casing can lead to premature and unwarranted failure of the casing, auger or center core.

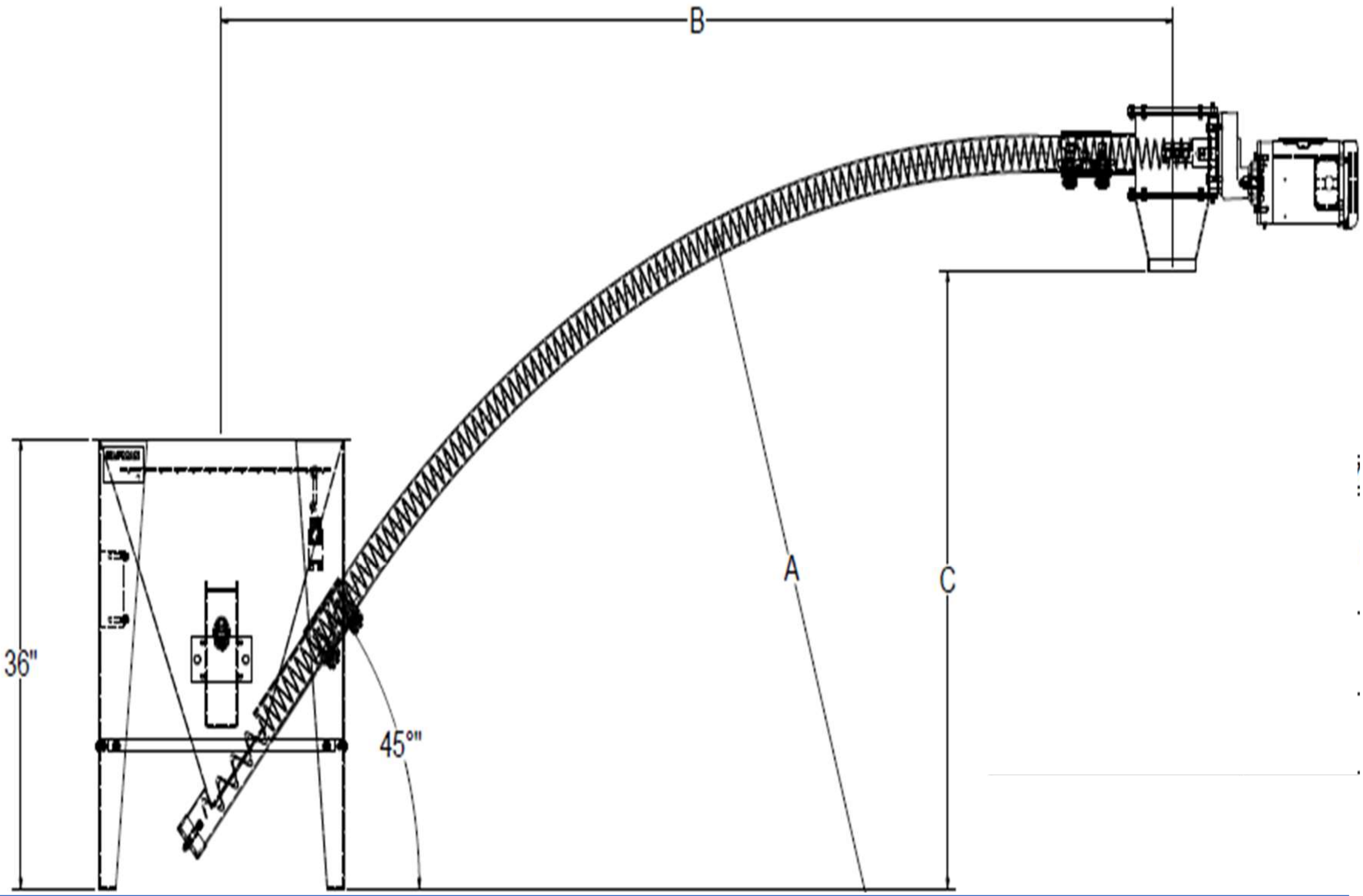






8.0 Installation: Helix Casing

If it is necessary for the auger/casing to be curved, maintain a smooth and generous radius to prevent a collapse of auger/casing wall. A tight radius will cause premature wear on auger and casing.

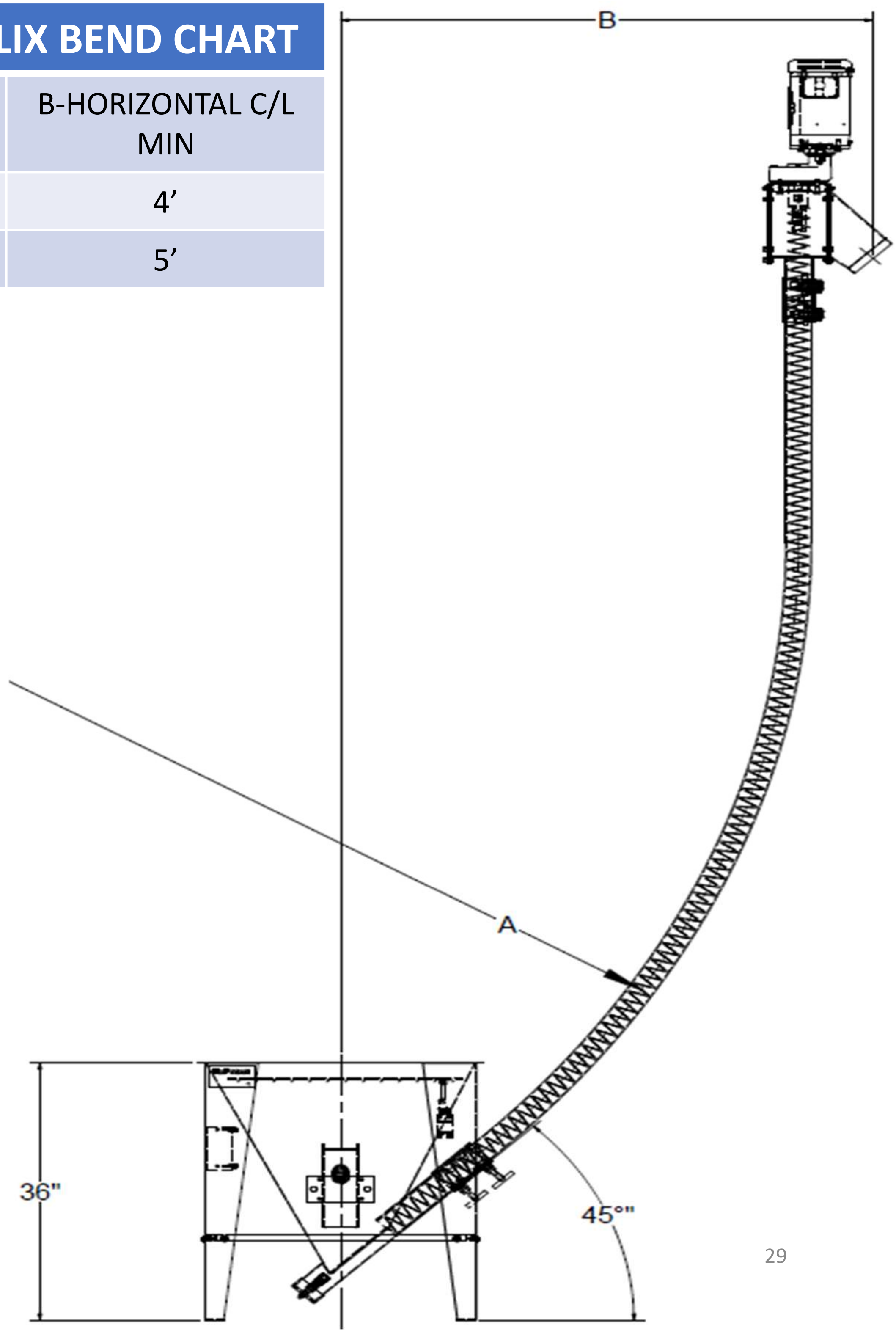


HORIZONTAL FLEXIBLE HELIX BEND CHART

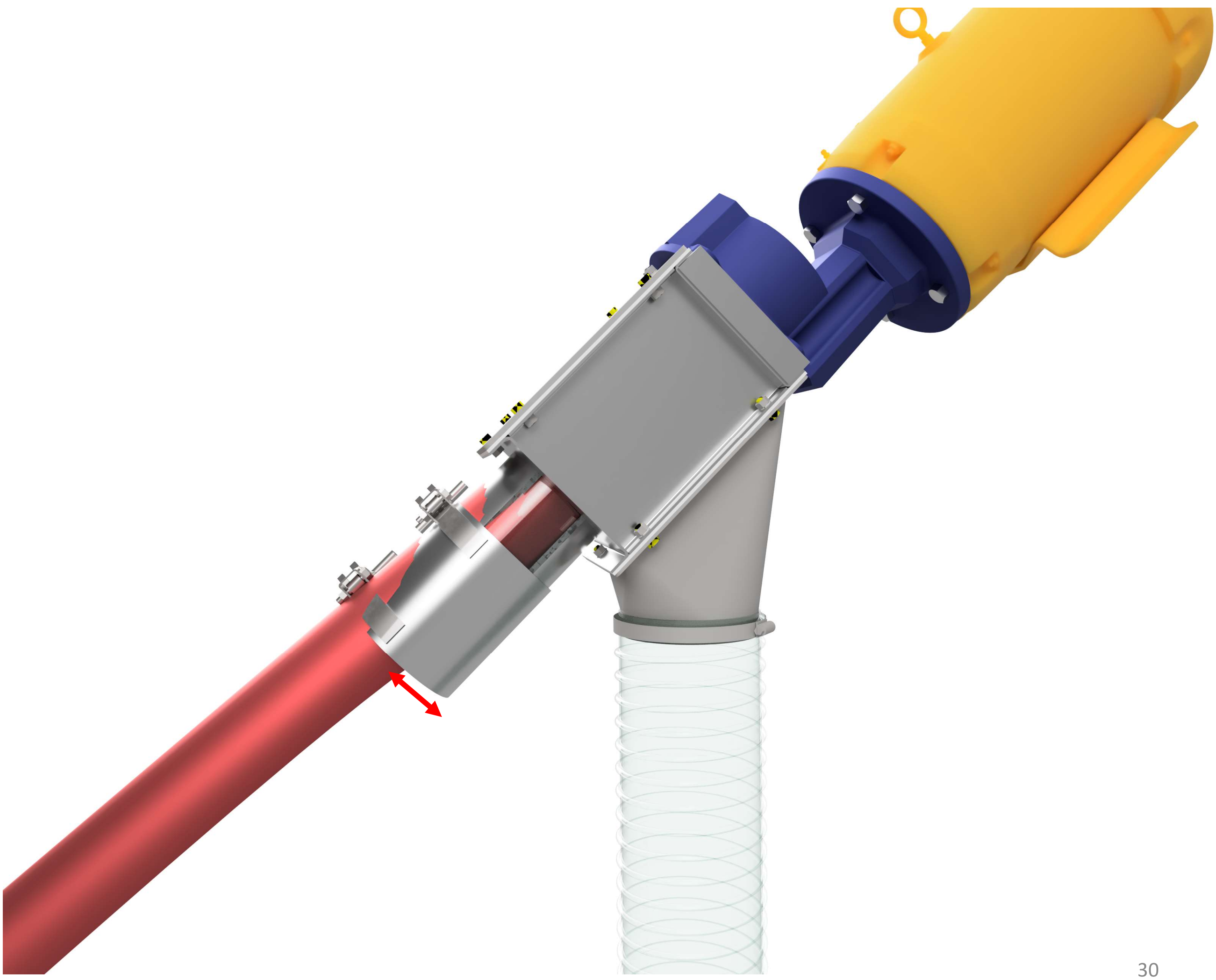
HELIX AUGER SIZE	A-BEND RADIUS MIN.	B-HORIZONTAL C/L MIN	C-VERTICAL C/L MIN
#250	6’	7’	3’
#300	10’	10’	4’

8.0 Installation: Helix Casing

VERITCAL FLEXIBLE HELIX BEND CHART		
HELIX AUGER SIZE	A-BEND RADIUS	B-HORIZONTAL C/L MIN
250	6’	4’
300	10’	5’

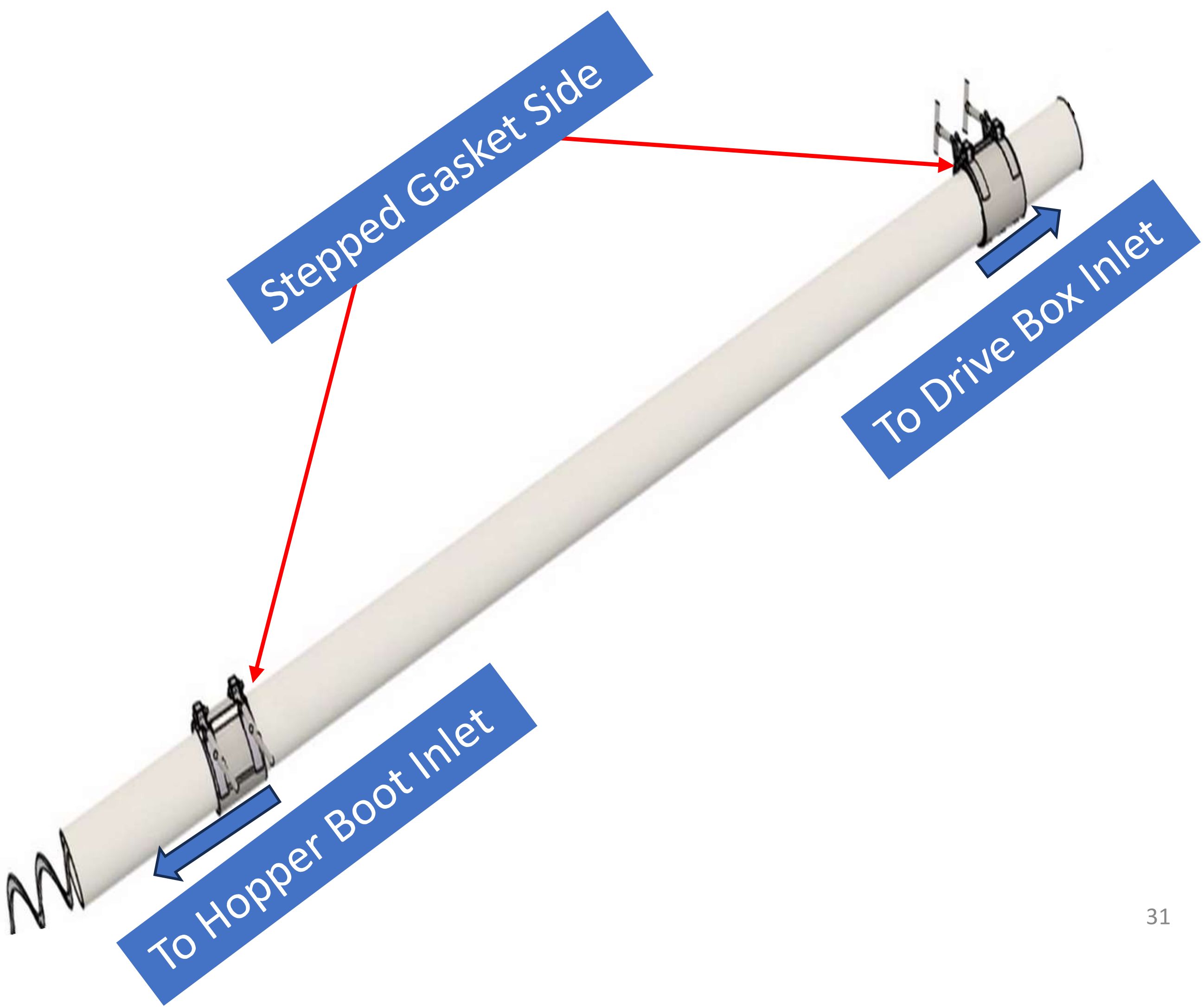


8.0 Installation: Casing (Drive Side)	
NOTICE	*A casing exerting excessive strain on the drive assembly or in-feed hopper may lead to premature wear or failure of Helix™ components.*
The image below shows a casing not aligned properly and pushing against the drive box. The casing must be in alignment with the drive and hopper. Do not force into place.	



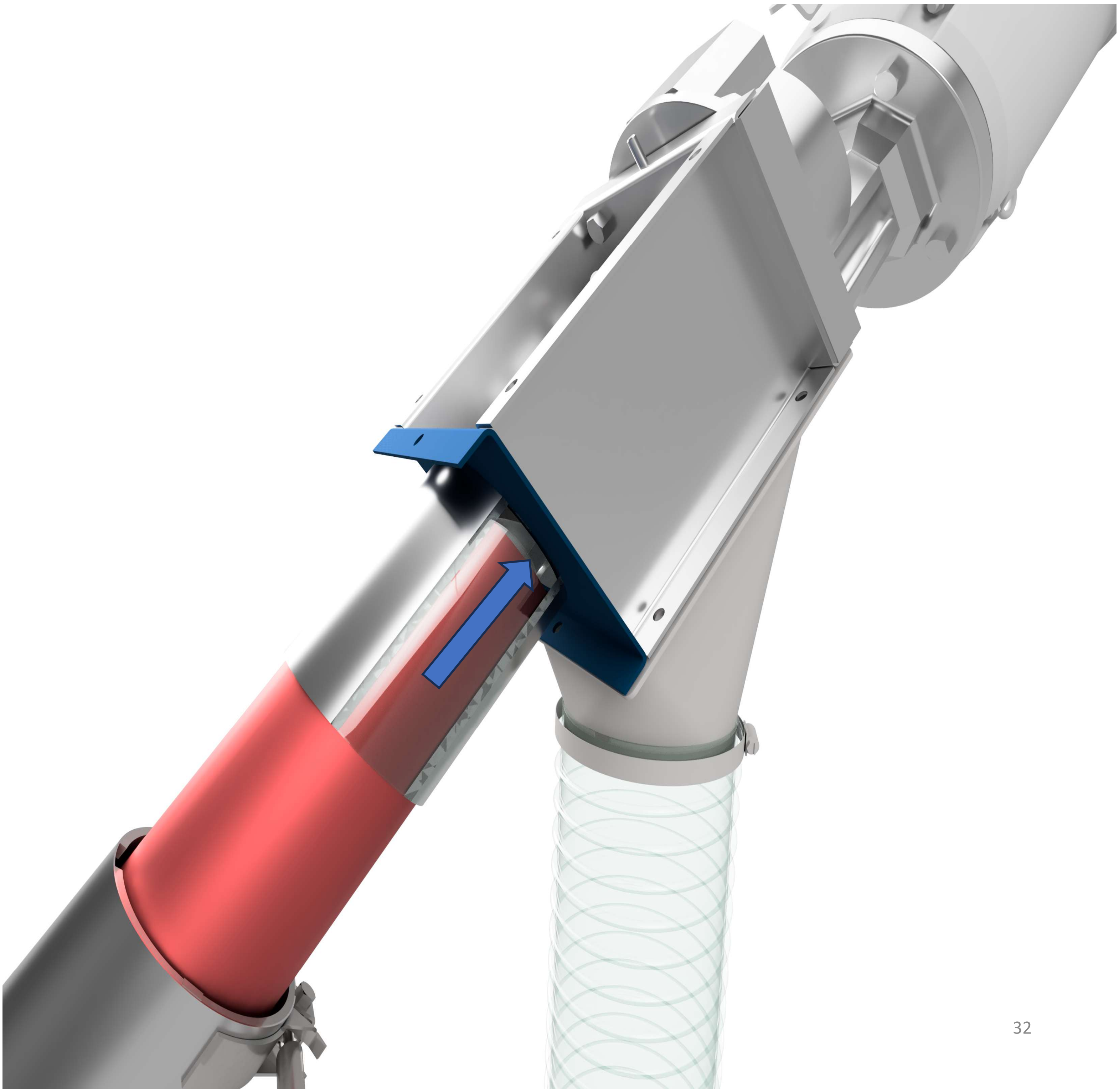
8.0 Installation: Casing (Drive Side)

NOTICE	*If you are installing a steel casing refer to Appendix G for the proper installation steps.*
1.	Inspect casing to make sure ends are cut square and straight. If using an existing casing, make sure the end being inserted to the drive box is cut straight and square.
2.	Slide stepped compression coupling onto casing and slide down a few inches from the end of the casing. The stepped side of the coupling should be on the casing. The straight side of the coupling should be facing the drive box and the hopper boot inlet.



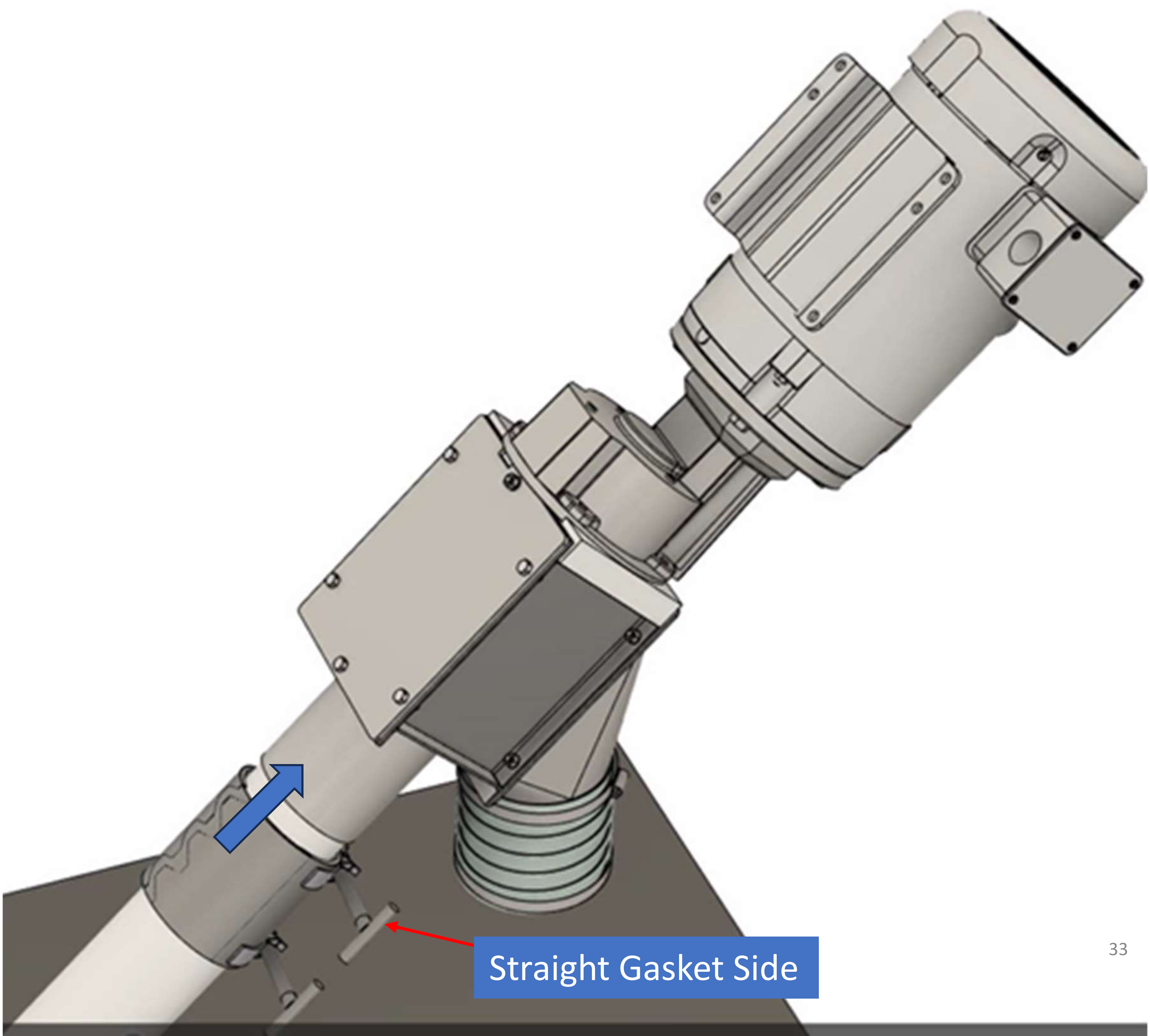
8.0 Installation: Casing (Drive Side)

3. Slide the casing (shown in red) into drive box until it is fully seated against the front drive box plate(shown in blue).



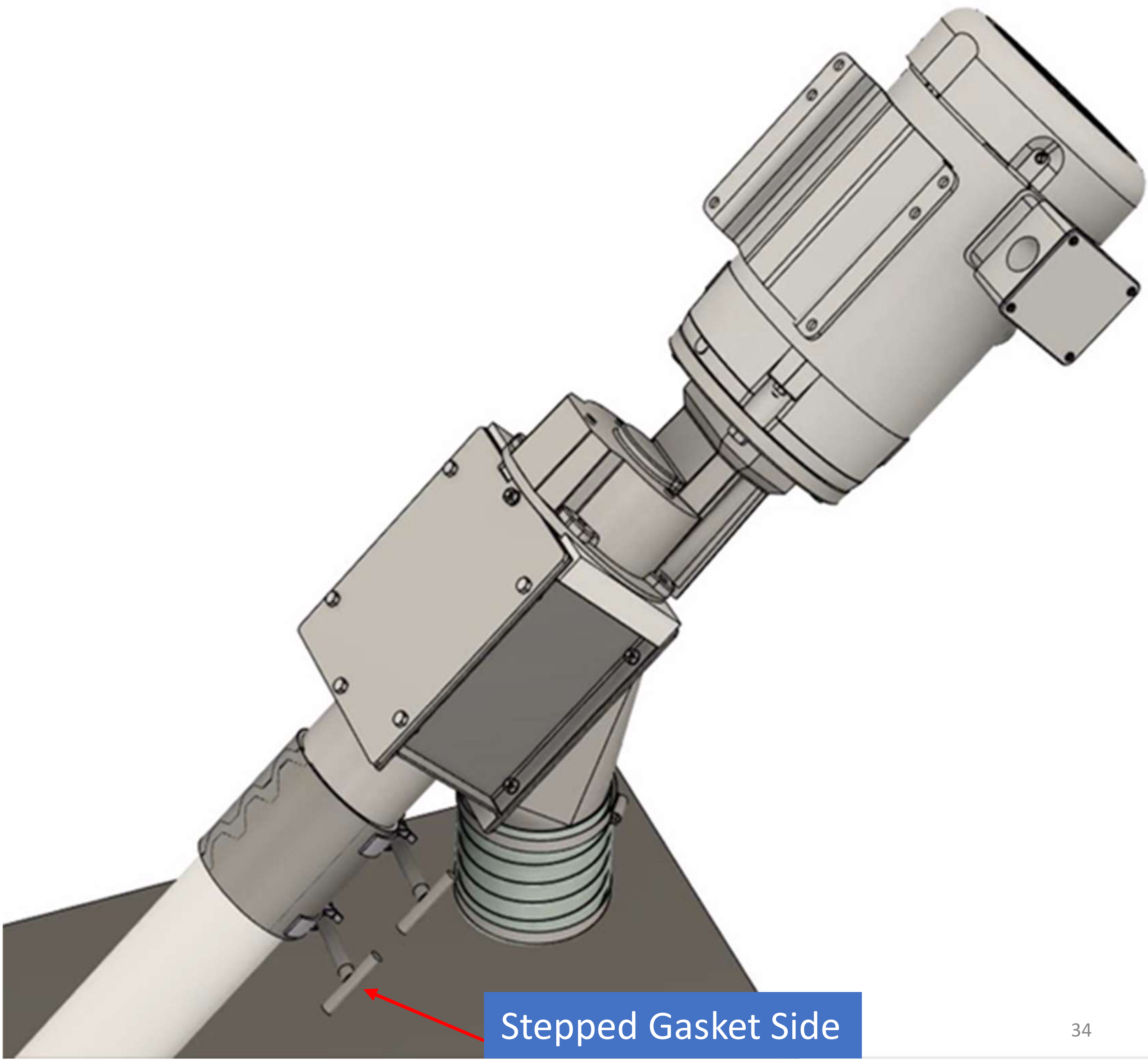
8.0 Installation: Casing (Drive Side)

4. Now slide the compression coupling up until the drive box inlet touches the stepped gasket. Then tighten down the straight gasket side of the clamp touching the metal drive box inlet until it is secure.



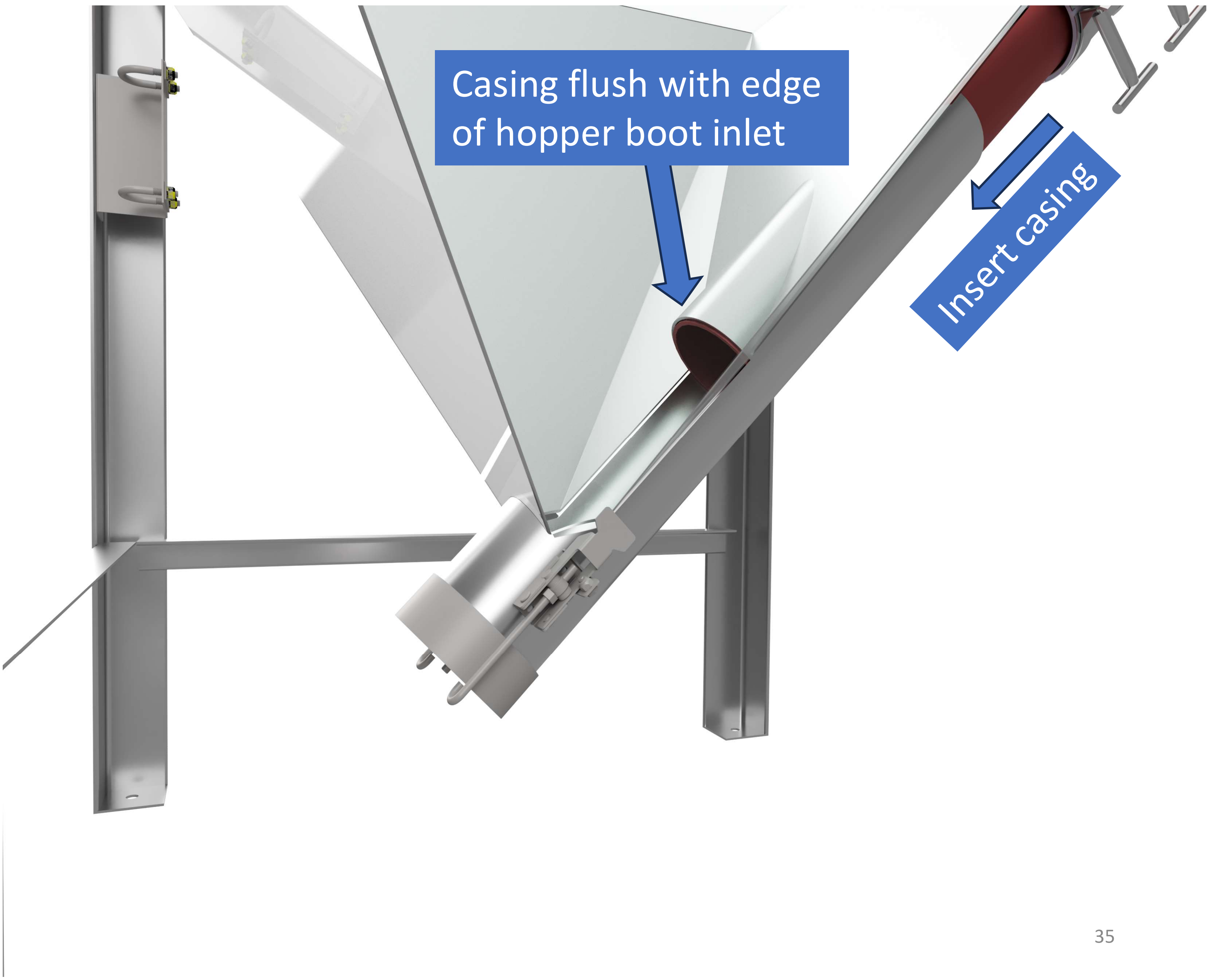
8.0 Installation: Casing (Drive Side)

5.	Now tighten down the stepped gasket side of the coupling to the Helix™ casing until clamp is secure.
NOTICE	*Do not over tighten stepped side of the coupling as this can compromise the casing and lead to premature failure.*



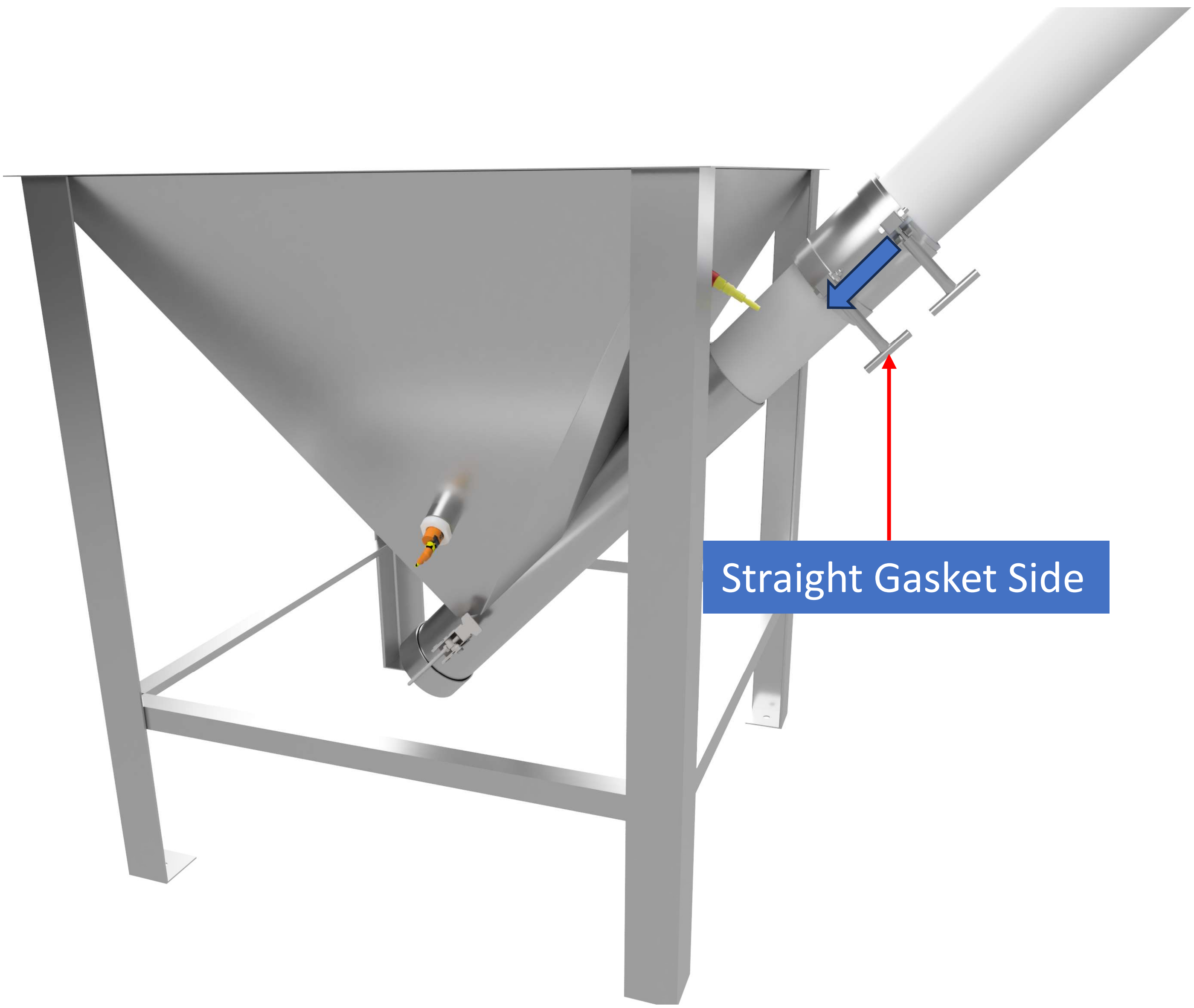
8.0 Installation: Casing (Hopper Side)

1.	Slide stepped compression coupling onto casing and slide it a few inches from the end of the casing. (The stepped side of the coupling should on the casing. The straight side of the coupling should be facing the Helix™ hopper inlet boot)
2.	Slide casing (shown in red) into the hopper until it is flush with the hopper boot inlet.



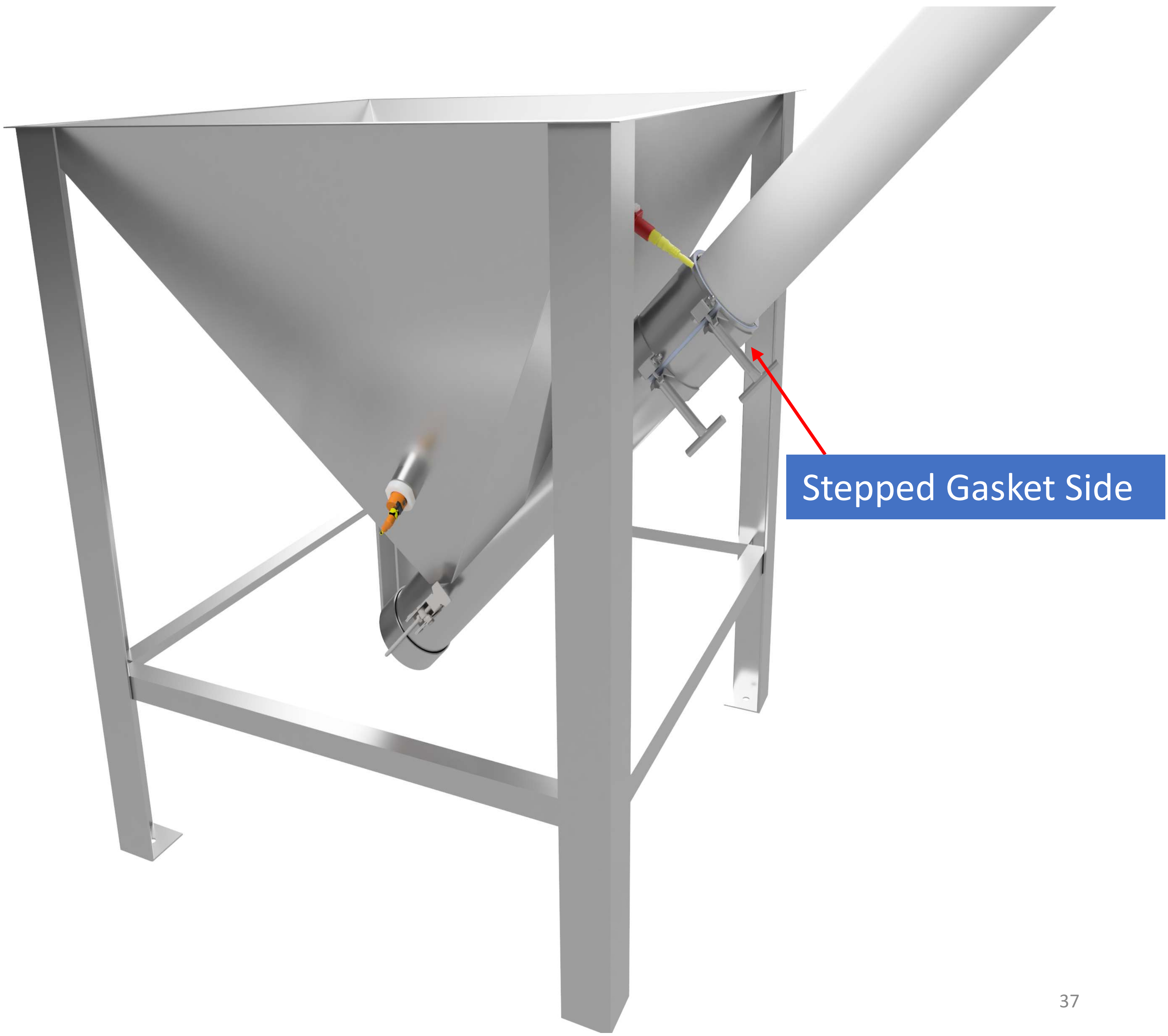
8.0 Installation: Casing (Hopper Side)

3.	Now slide the compression coupling down until the hopper’s boot inlet is seated in the straight gasket side of the coupling. Then tighten down the straight gasket side of the clamp to the metal hopper boot inlet until clamp is secure.

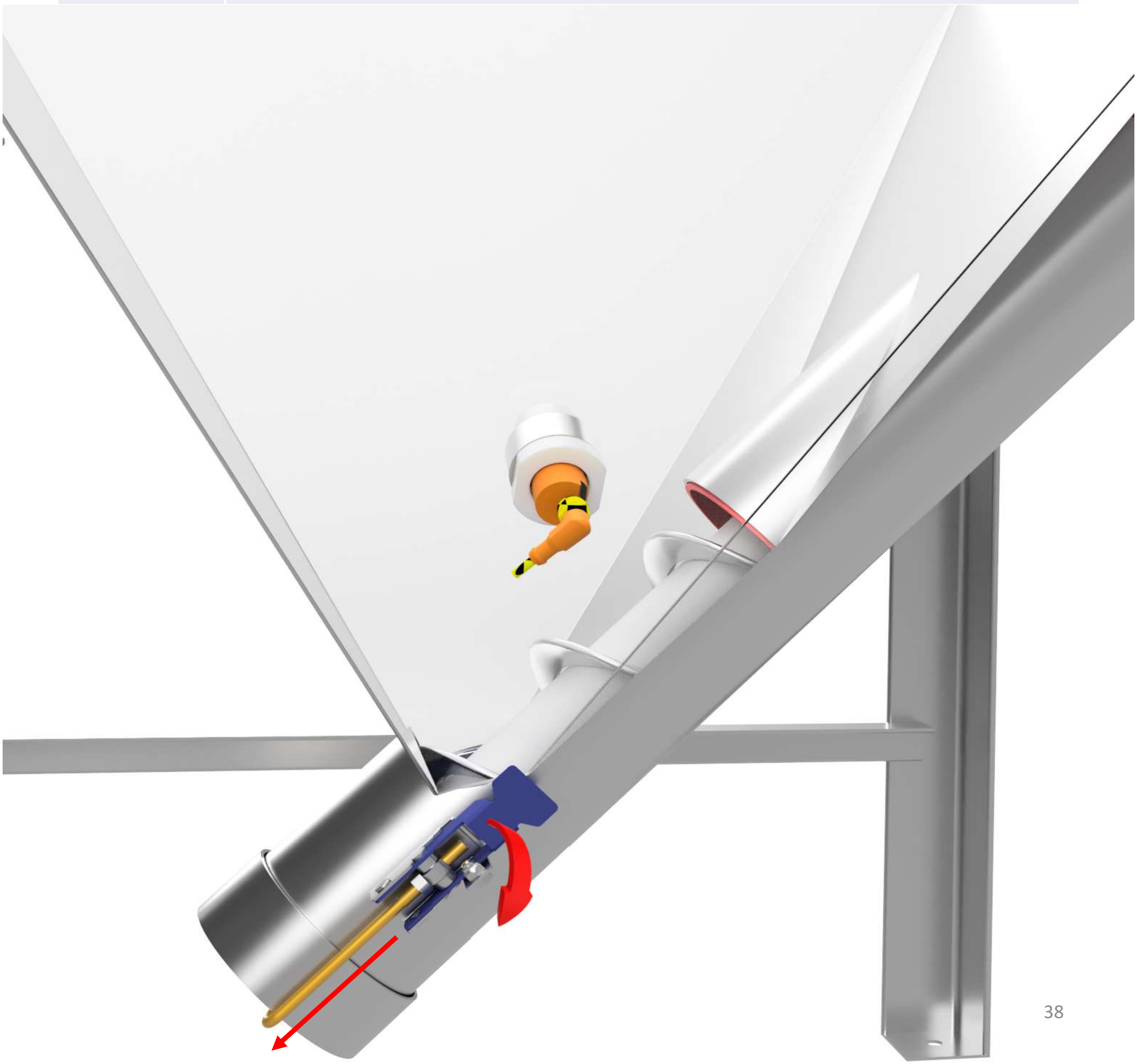


8.0 Installation: Casing (Hopper Side)

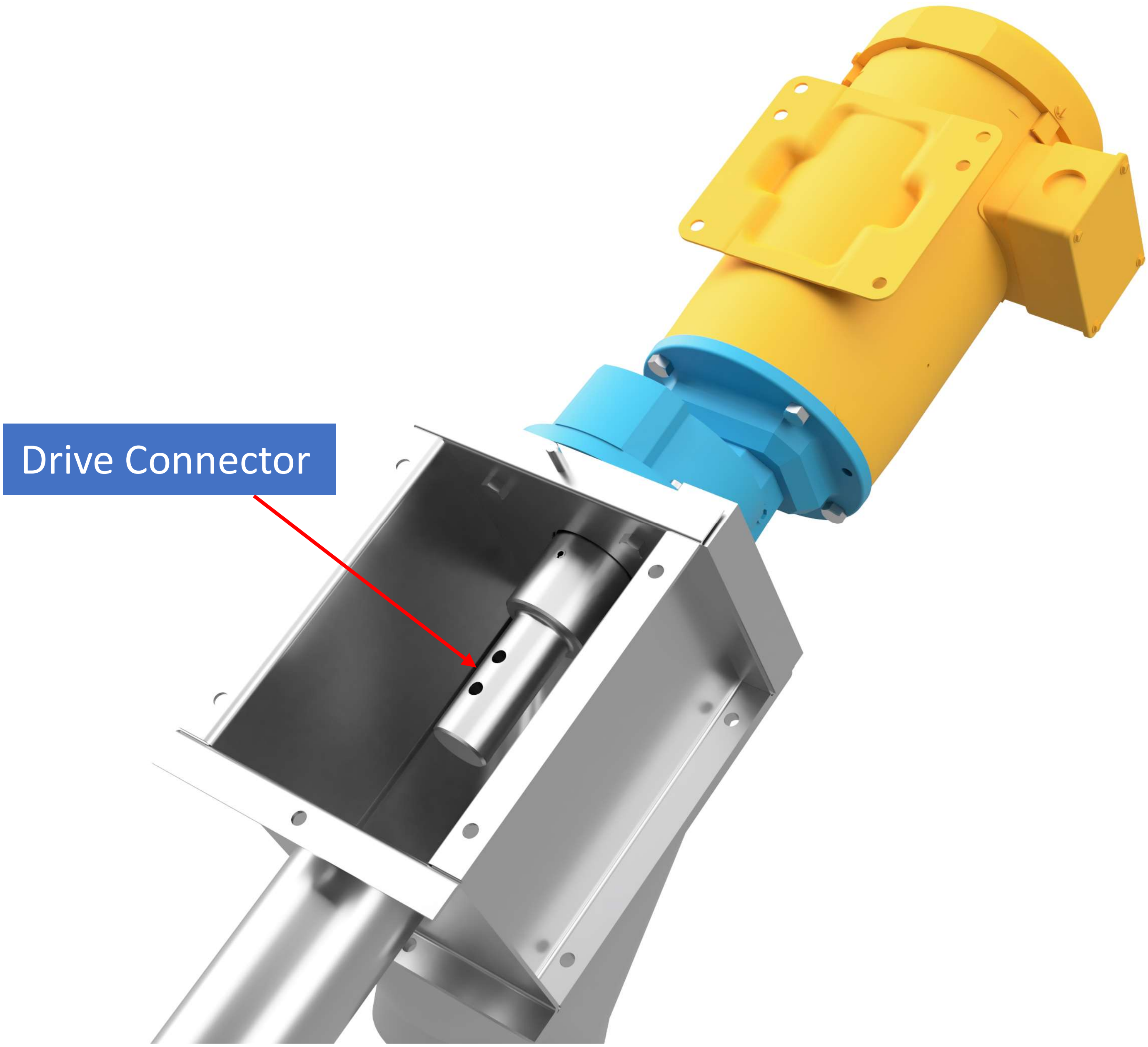
4.	Now tighten down the stepped gasket side of the coupling to the Helix casing until clamp is secure.
NOTICE	Do not over tighten stepped side of the coupling as this can compromise the casing and lead to premature failure.



8.0 Installation: Helix™ Auger	
1.	Ensure all power is locked out at main panel and any local disconnects during the entire installation process.
2.	Remove end cap from hopper boot by releasing the j-hook latches and pulling off cap.



8.0 Installation: Helix™ Auger	
3.	Remove the top cover of the drive box so you can see the drive connector inside.



8.0 Installation: Helix™ Auger

4.	Confirm the open end of the auger is facing toward the drive box.
NOTICE	**If installing a beveled edge auger reference Appendix B before starting installation.**

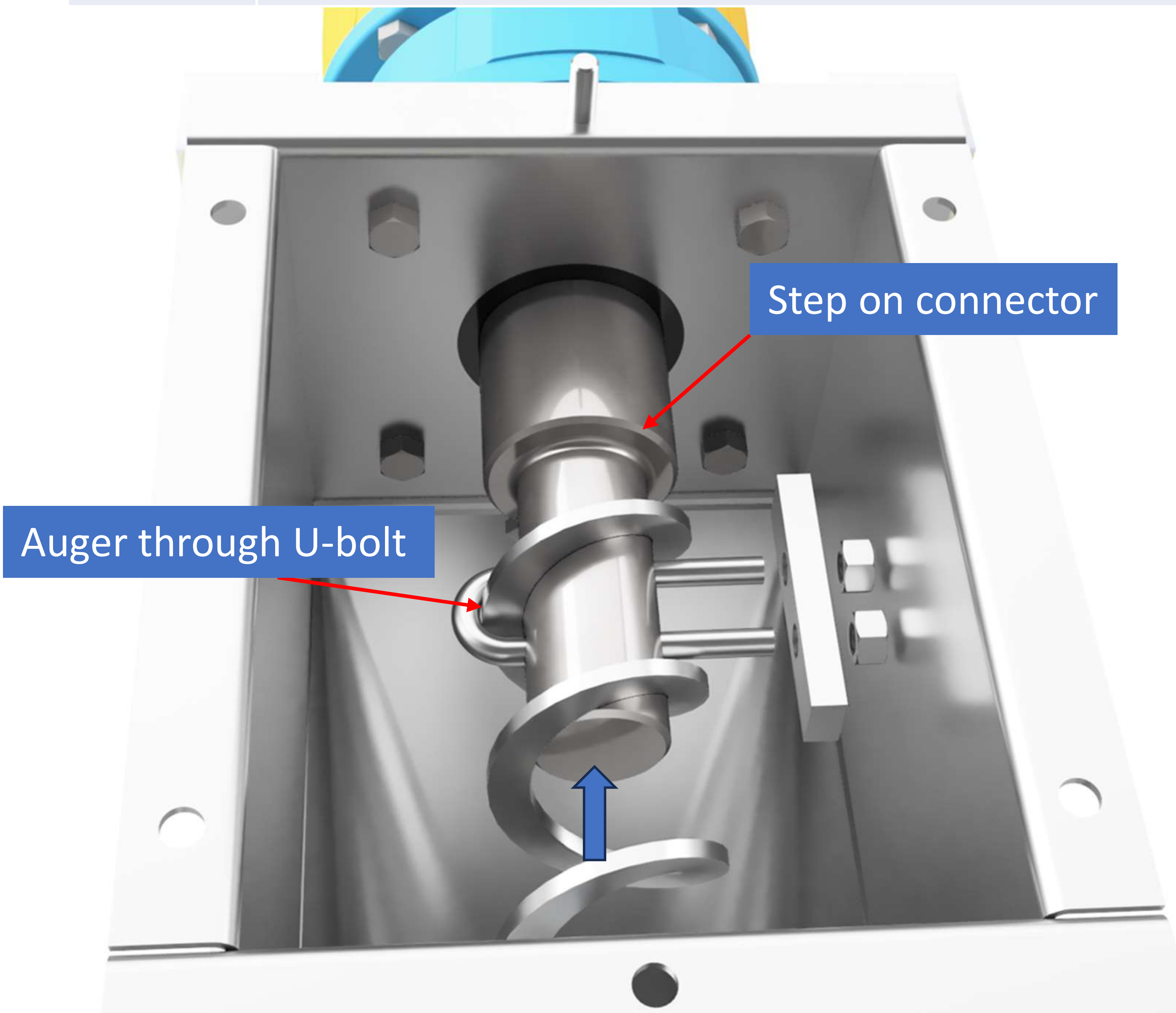
Open End of Auger	Closed End of Auger
	

8.0 Installation: Helix™ Auger

5.	Insert open end of the auger into the hopper boot and push up towards the drive box

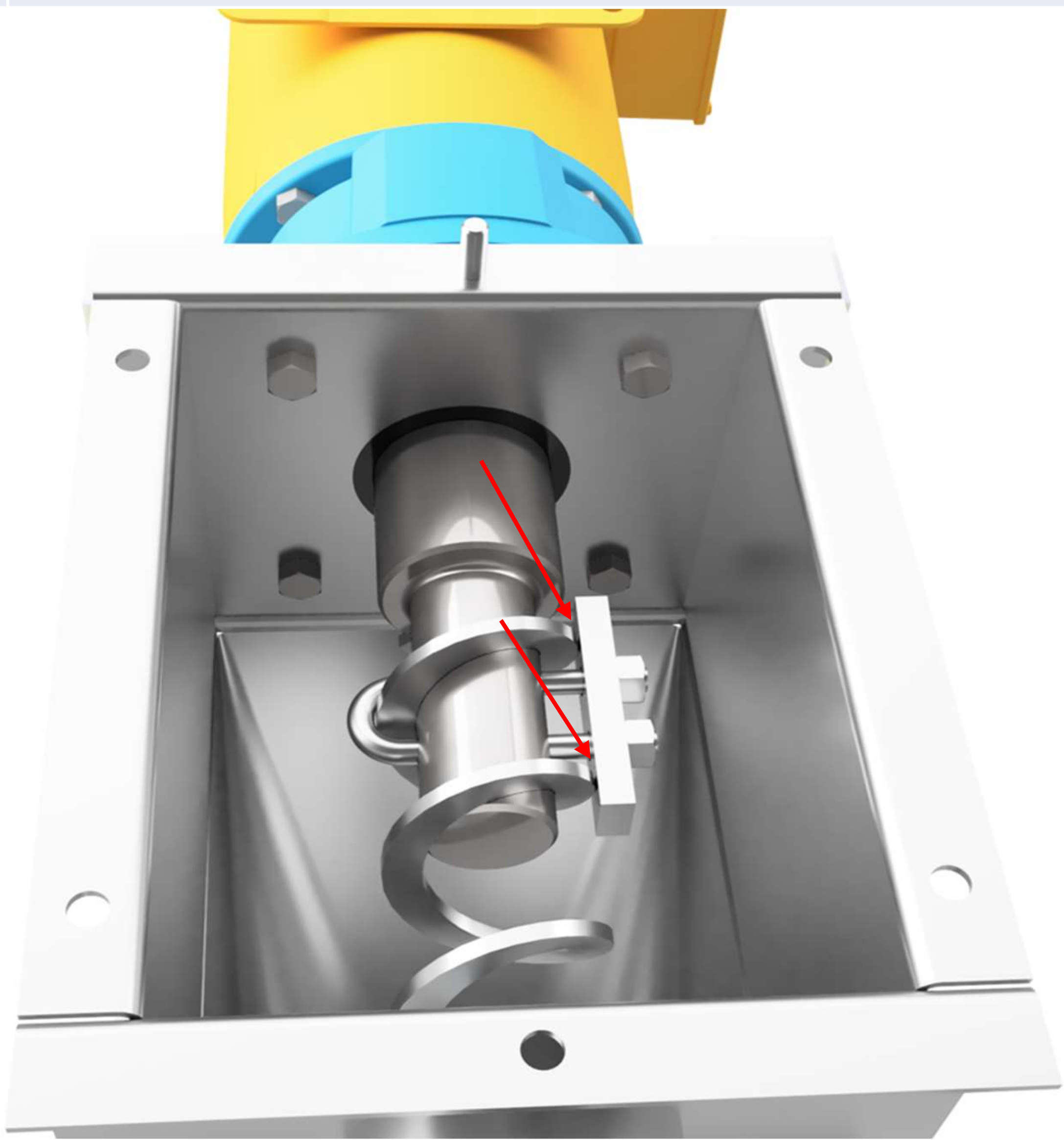
8.0 Installation: Helix™ Auger

6.	Slide Helix™ auger over the drive connector until auger contacts the step in the connector.
7.	Insert U-Bolt through connector holes. Ensure that a flight of the auger is centered in the U-Bolt.

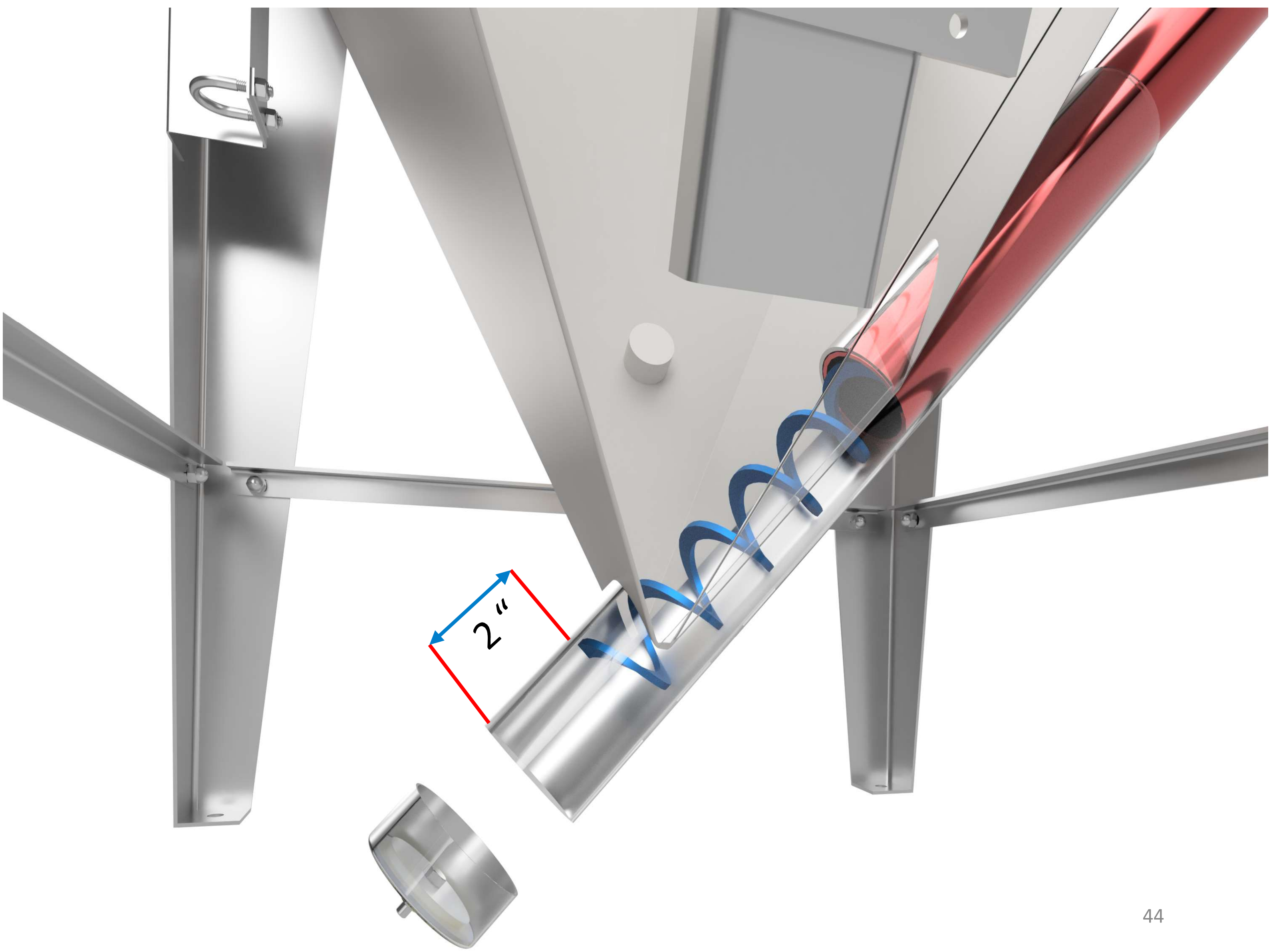


8.0 Installation: Helix™ Auger

8.	Slide the retaining bar onto the end of the U-bolt. ***(This bar should be <u>spanning across two flights</u> as depicted in the image below.)***
9.	Install Nyloc nuts and tighten down until snug. ***(Do not overtighten nuts as this can bend the retaining bar and cause the auger to come loose during operation.)***

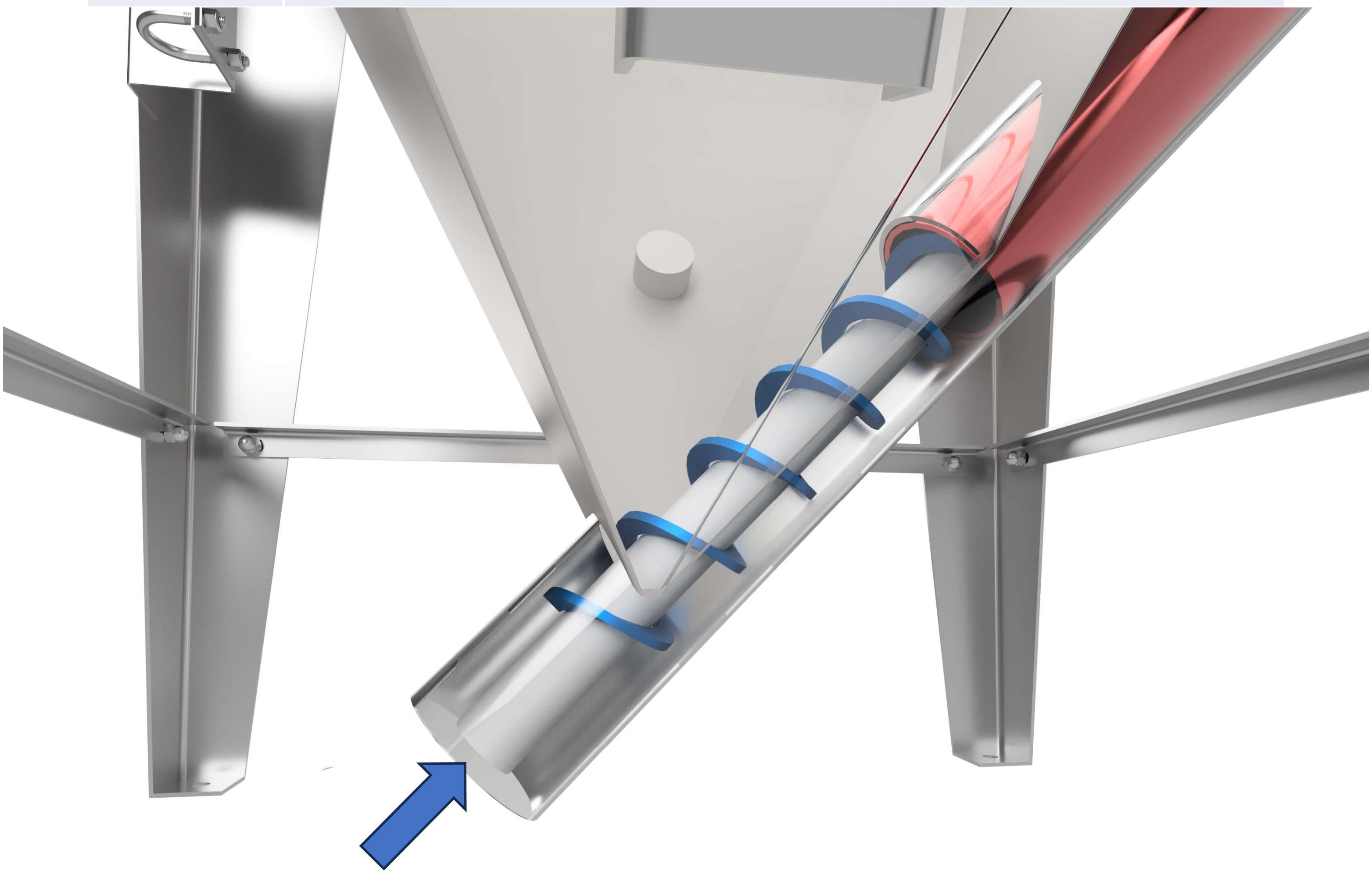


8.0 Installation: Helix™ Auger	
10.	Once secured in place, go to the hopper boot and verify that there is 2 inches of space between the tail of the auger and the end cap.
NOTICE	*If the gap is less than 2 inches, then you will need to remove the auger and trim it. Reference Appendix A for trimming procedure.*
NOTICE	*If the auger is cut too short, it will not recess properly in the boot. This could lead to the end of the auger whipping around causing damage to the hopper, auger, and casing.*



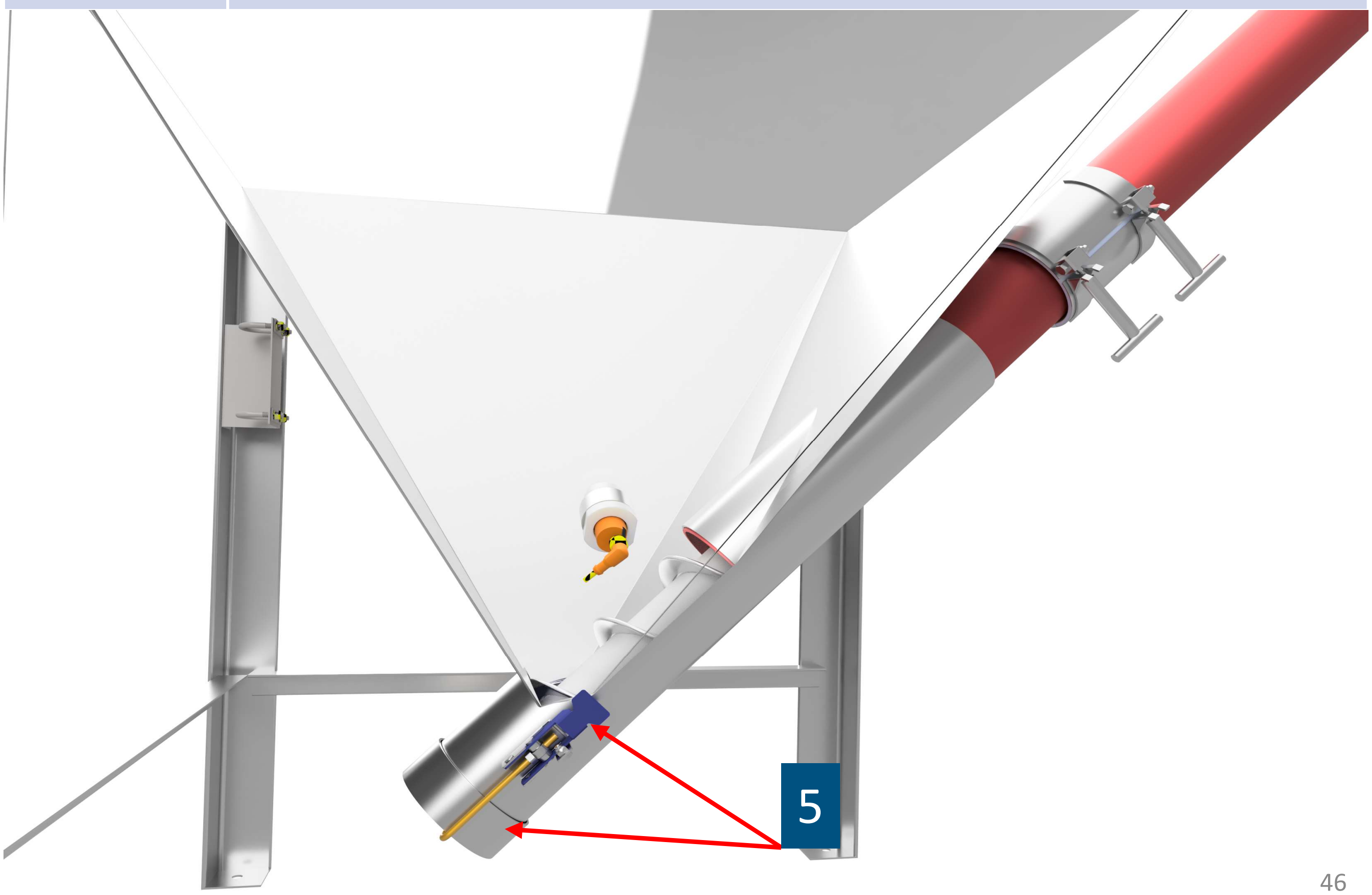
8.0 Installation: Center Core

1.	Ensure all power is locked out at main panel and any local disconnects
2.	Remove end cap from hopper boot
3.	Insert center core into center of the auger at the hopper boot and push up towards the drive box
NOTICE	***The center core is intended to float inside the ID of the auger so it will be shorter in length than the auger.***



8.0 Installation: Center Core

4.	Verify that the center core did not push up into the auger flighting in the drive box and is contacting the face of the drive connector.
5.	Install end cap back on and secure in place using the latches.
NOTICE	*If the end cap is not properly secured it could come off during operation.*



9.0: Mechanical Installation Verifications

1.	Check that Helix™ auger has been attached and secured properly (via U-bolt or J-bolts) to the connector in the drive box.
2.	Inspect casing connection points to the hopper and drive box. Verify that the compression couplings are installed in the correct direction and have been tightened down properly.
3.	Inspect hopper boot with end cap removed and verify that the Helix auger is the correct distance from the end cap.
4.	Verify that the gear box has the oil fill at the proper level, and that the gear box breather vent is installed.
NOTICE	*The breather vent should be at the highest point on the gearbox. If you have any questions, please contact our service department.*
5.	Visually inspect to make sure there are no tools, debris, etc. in the conveyor from installation process.
6.	Check that all guards and covers are properly mounted and secured.